

ARACHIS ONLINE TRAINING

COMPLETE BEGINNER'S GUIDE TO MAKING METHYLATED SPIRIT

Professional Practical Training Manual

INTRODUCTION

Welcome to Arachis Online Training.

In this lesson, you will learn how to make ****Methylated Spirit**** step by step using practical methods suitable for small-scale production in Zimbabwe.

This training is designed for:

- * complete beginners
- * students starting small businesses
- * tuckshop suppliers
- * cleaning chemical manufacturers
- * industrial product resellers

By the end of this lesson, you should understand:

- * what methylated spirit is
- * ingredients used
- * functions of each ingredient
- * mixing procedures
- * safety precautions
- * packaging
- * costing basics
- * selling strategies

WHAT IS METHYLATED SPIRIT?

Methylated spirit is a denatured alcohol product mainly used for:

- * cleaning surfaces
- * disinfecting
- * stain removal
- * laboratory use
- * spirit burners
- * fuel burners
- * industrial cleaning

It is called “methylated” because chemicals are added to alcohol to make it poisonous and impossible to drink.

The product usually appears:

- * purple
- * blue
- * clear blue

depending on the dye used.

UNDERSTANDING THE MAIN INGREDIENTS

Before making the product, you must understand what each ingredient does.

1. ETHANOL

What Is Ethanol?

Ethanol is the main alcohol used in methylated spirit.

It:

- * evaporates quickly
- * burns easily
- * kills germs
- * removes dirt and grease

It is the main active ingredient.

Recommended Strength

Best results come from:

- * 95% ethanol or higher

Lower-quality ethanol reduces performance.

Function of Ethanol

Ethanol provides:

- * cleaning power
- * fast drying
- * burning capability

- * disinfecting ability

Without ethanol, the product will not work properly.

2. METHANOL

What Is Methanol?

Methanol is another type of alcohol added to make the product unsafe for drinking.

This process is called:

DENATURING

Why Methanol Is Added

Methanol:

- * prevents human consumption
- * changes the smell
- * improves burning characteristics
- * helps comply with regulations

IMPORTANT WARNING

⚠ Methanol is highly poisonous.

If swallowed, it can:

- * damage eyesight
- * cause blindness
- * cause death

Never taste the product.

Never inhale heavily.

Always work in ventilated areas.

3. WATER

Why Water Is Added

Water helps:

- * reduce alcohol strength slightly
- * control evaporation rate
- * reduce production costs
- * improve burner performance

What Type of Water Should Be Used?

Best:

- * distilled water

Acceptable:

- * clean borehole water
- * filtered water

Avoid:

- * dirty water
- * salty water
- * muddy water

Dirty water causes:

- * cloudiness
- * contamination
- * reduced quality

4. DYE

Why Dye Is Added

Dye helps identify the product as methylated spirit.

Common colors:

- * purple
- * blue

This prevents accidental drinking.

Common Dyes

You may use:

- * methyl violet
- * blue dye

Only small amounts are needed.

Too much dye:

- * stains surfaces
- * makes product look poor quality

5. BITREX (OPTIONAL)

What Is Bitrex?

Bitrex is an extremely bitter chemical.

It is added to:

- * discourage drinking
- * improve safety

Only a very small amount is needed.

COMPLETE FORMULA (20 LITRES)

Ingredient	Quantity
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Ethanol (95%+)	18 Litres
Methanol	1 Litre
Clean Water	900ml
Dye	Small amount
Bitrex (Optional)	5–10ml

EQUIPMENT REQUIRED

Equipment	Purpose
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20L bucket/drum	Mixing
Measuring jug	Accurate measurement

Stirring stick	Mixing	
Funnel	Filling bottles	
Gloves	Safety	
Face mask	Fume protection	
Bottles	Packaging	

STEP-BY-STEP PROCEDURE

STEP 1 — PREPARE YOUR WORKING AREA

Before mixing:

- * work in open air or ventilated room
- * remove all flames
- * remove cigarettes
- * remove sparks

Alcohol products catch fire very easily.

STEP 2 — WEAR SAFETY EQUIPMENT

Wear:

- * gloves
- * mask
- * protective clothing

This protects your skin and lungs.

STEP 3 — ADD ETHANOL

Pour:

- * 18 litres ethanol

into the mixing container.

This forms the main base of the product.

STEP 4 — ADD METHANOL

Slowly add:

* 1 litre methanol

while stirring gently.

Do not splash.

Avoid breathing fumes directly.

STEP 5 — ADD WATER

Add:

* 900ml clean water

and continue stirring.

Water slightly reduces strength while maintaining effectiveness.

STEP 6 — ADD DYE

Add a small amount of dye.

Start with:

* tiny pinches for powder dye

OR

* a few drops for liquid dye

Mix thoroughly.

STEP 7 — ADD BITREX (OPTIONAL)

Add:

* 5–10ml Bitrex

and mix well.

Remember:

too much Bitrex can create an unpleasant smell.

STEP 8 — MIX THOROUGHLY

Stir continuously for:

- * 5–10 minutes

until fully uniform.

The product should appear:

- * clean
- * evenly colored
- * free from particles

STEP 9 — PACKAGE THE PRODUCT

Use:

- * strong plastic bottles
- * tightly sealing caps

Common sizes:

- * 500ml
- * 1 litre
- * 2 litres
- * 5 litres

HOW TO CALCULATE ALCOHOL STRENGTH

Approximate final alcohol strength:

90%

This is strong enough for:

- * cleaning
- * disinfecting
- * burners

QUALITY CHECKS

A good methylated spirit should:

- * dry quickly
- * burn properly
- * smell alcoholic
- * appear clean
- * have consistent color

COMMON MISTAKES BEGINNERS MAKE

1. Adding Too Much Water

Result:

- * weak product
- * poor burning
- * slow drying

2. Using Dirty Water

Result:

- * cloudy product
- * contamination

3. Too Much Dye

Result:

- * staining
- * unprofessional appearance

4. Mixing Near Fire

Result:

- * explosions
- * fire accidents

SAFETY RULES

⚠ Highly flammable

- ⚠ Never smoke while mixing
- ⚠ Keep away from children
- ⚠ Store in cool places
- ⚠ Avoid inhaling fumes
- ⚠ Never drink the product

STORAGE

Store:

- * tightly sealed
- * away from sunlight
- * away from heat

Poor storage causes:

- * evaporation
- * reduced strength

FINAL ADVICE TO STUDENTS

Start small.

Focus on:

- * consistency
- * cleanliness
- * safety
- * proper packaging

As your confidence grows:

- * increase production
- * improve branding
- * expand distribution

The chemical business rewards:

- * discipline
- * quality
- * consistency

END OF TRAINING MODULE

ARACHIS ONLINE TRAINING

“Practical Skills for Real Business”